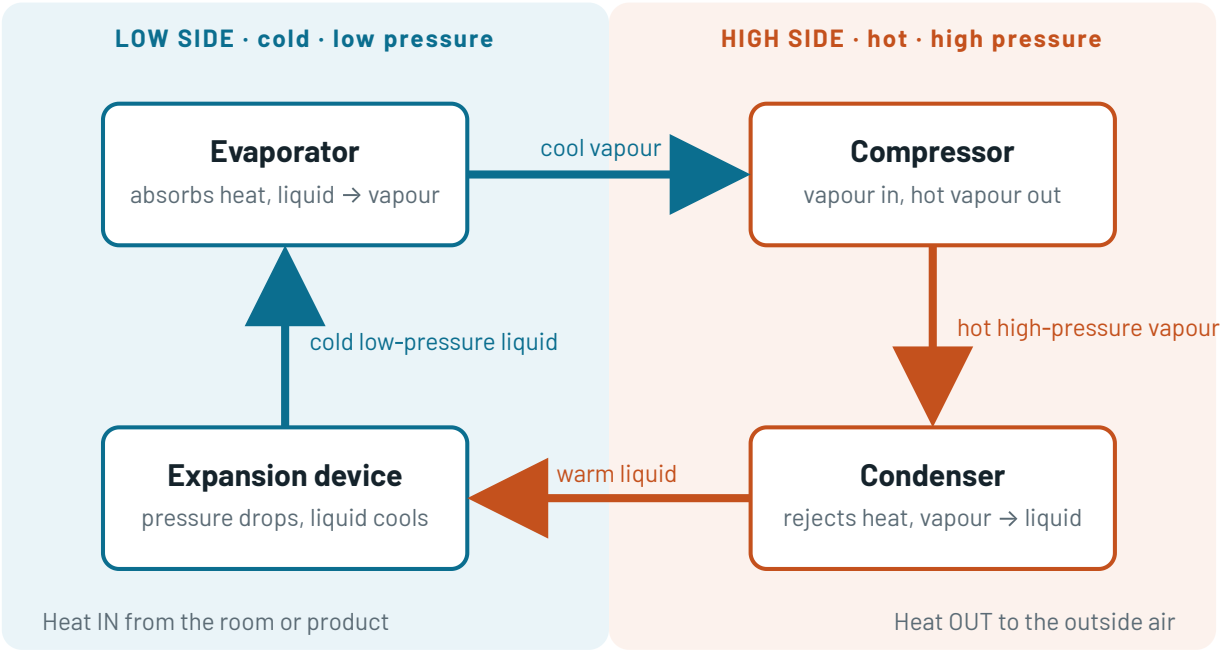
CITY & GUILDS 2079-11 · LEVEL 2 AWARD · F-GAS & ODS REGULATIONS · CATEGORY I

F-Gas Category I Study Guide

The refrigeration cycle, refrigerants, the law, leak checking, recovery, commissioning and safety, explained in pictures, with every number you need on one page and 30 practice questions.



Great Britain edition · 2026 · Written for the GB F-Gas Regulation (retained EU 517/2014) and the ODS Regulation

Contents

1 **How the exam works**
What City & Guilds 2079-11 actually tests, and how to use this guide alongside the practice app.

2 **The refrigeration cycle in pictures**
Four components, two pressures, one job: move heat from where it is not wanted to where it does no harm.

3 **Refrigerants**
Families, numbers, GWP, blends, safety classes and oils. Half the exam lives in this chapter.

4 **The law: F-Gas and ODS**
Who may do what, what the operator must do, and the ban dates that keep coming up.

5 **CO2 equivalent and leak checking**
One formula, three thresholds, and the methods the assessor wants to see.

6 **Recovery and disposal**
Getting refrigerant out safely, keeping it clean, and moving it on legally.

7 **Installation and commissioning**
Test, evacuate, charge, run. In that order, every time.

8 **Reading the symptoms**
Four faults account for most call-outs. Learn their fingerprints.

9 **Health and safety**
Refrigerant is cold, heavy, under pressure, sometimes flammable, and poisonous in a flame.

10 **Records**
Five years, both parties, and exactly what goes in.

11 **Practice questions**
30 exam-style questions with an answer key and explanations.

12 **Numbers to memorise**
Every threshold, date and value on one page.

13 **Glossary**
94 terms in plain English.

Companion practice app: **fgas-trainer.vercel.app**. The questions and glossary in this guide are the same ones the app uses, so they always agree.

This is independent revision material written from the published syllabus and the GB regulations, using the GWP values the regulation uses (IPCC AR4). It is not official City & Guilds content and is not affiliated with City & Guilds or any training provider. Where your course workbook or tutor differs, follow them.

1 How the exam works

What City & Guilds 2079-11 actually tests, and how to use this guide alongside the practice app.

What 2079-11 is

The Level 2 Award in F-Gas and ODS Regulations, Category I, is the qualification that lets you legally work on any stationary refrigeration, air conditioning or heat pump system containing fluorinated greenhouse gases, whatever its charge size. It is the full ticket. Categories II, III and IV are cut-down versions limited by activity or charge.

The course is normally one or two days. Assessment is in two parts: a **multiple-choice theory test** on the regulations, refrigerants, the refrigeration cycle, leak checking, recovery and safety, and a **practical assessment** where you demonstrate leak checking, recovery, pressure testing, evacuation and charging on a rig, with the assessor asking questions as you go.

What the theory test rewards

Most questions are about **numbers and definitions**: the 5, 50 and 500 tonne thresholds, the 3 kg and 6 kg category limits, GWP values, 80% cylinder fill, 500 microns, 5 years of records. The rest ask you to apply them: given a charge and a refrigerant, how often is the leak check? The maths is one multiplication and one division.

Chapter 12 is a one-page list of every number you need. Learn that page and you have most of the marks.

How to use this guide

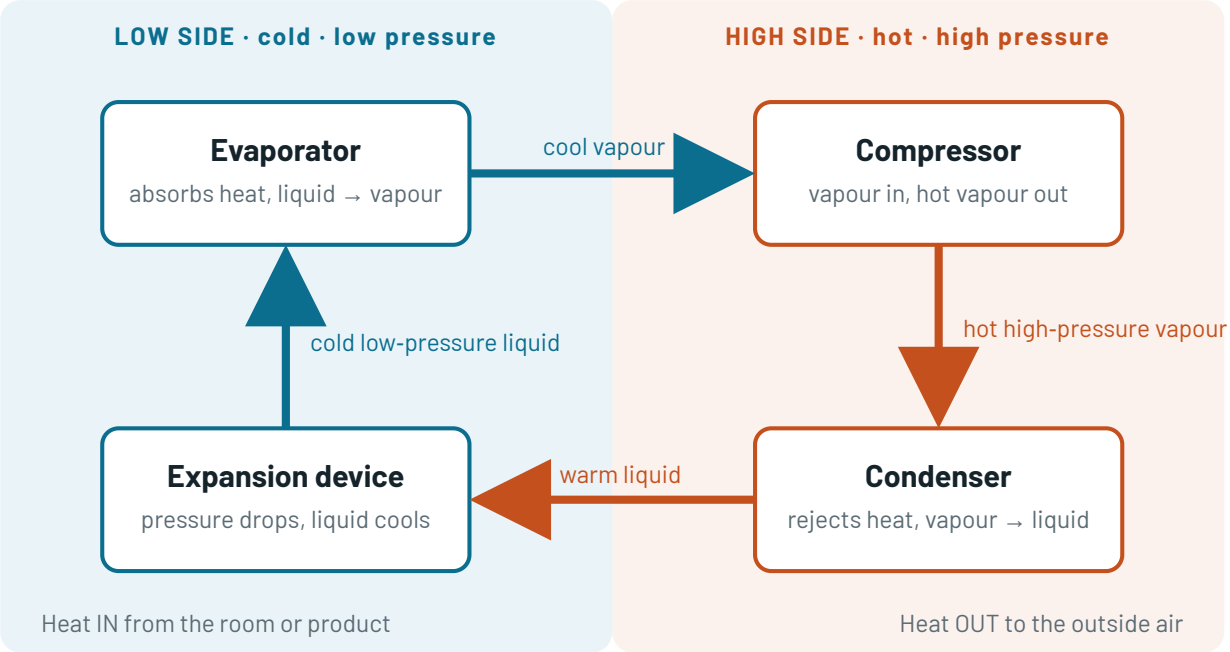
Read a chapter, then run that topic in the practice app. The app explains every wrong answer, and the dotted words open the same glossary printed at the back of this guide. Finish with two or three mock exams at 70% or better before test day.

> **Great Britain versus the EU.** GB kept EU Regulation 517/2014 as its own F-Gas Regulation after Brexit. The EU has since replaced it with Regulation 2024/573, which applies in Northern Ireland. The GB rules are what 2079-11 tests. Where an online resource quotes EU 2024/573 dates, treat them with care.

2 The refrigeration cycle in pictures

Four components, two pressures, one job: move heat from where it is not wanted to where it does no harm.

The four components



Refrigerant goes round a closed loop. The **compressor** squeezes cool vapour into hot, high-pressure vapour. In the **condenser** that vapour gives its heat to the outside air and turns to warm liquid. The **expansion device** drops the pressure sharply, so the liquid can now boil at a low temperature. In the **evaporator** it does exactly that, soaking up heat from the room or the product, and returns to the compressor as cool vapour.

Everything from the compressor outlet to the expansion device is the **high side**: hot, high pressure, read on the red gauge. Everything from the expansion device to the compressor inlet is the **low side**: cold, low pressure, read on the blue compound gauge.

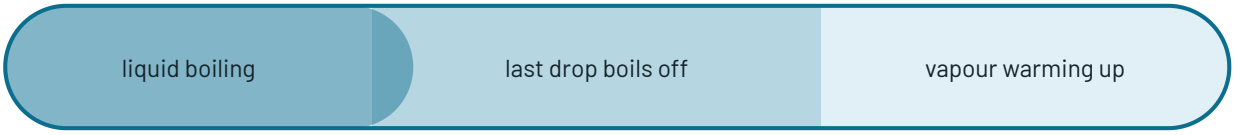
Why it gets cold: latent heat

Heat that changes the temperature of something is **sensible heat**. Heat that changes its state, liquid to vapour or back, without changing temperature is **latent heat**, and there is far more of it per kilogram. Refrigeration works by boiling liquid in the evaporator (absorbing latent heat) and condensing vapour in the condenser (releasing it).

A liquid boils at a temperature fixed by its pressure. Lower the pressure and it boils colder. That is all the expansion device does. For any refrigerant the relationship is printed on a **pressure-temperature (PT) chart**; reading a gauge pressure and converting it to a saturation temperature is the core skill of commissioning and fault finding.

Superheat and subcooling

Superheat is measured here



Evaporator → suction line. Saturation temperature from the suction pressure, actual temperature from a probe on the pipe.

Superheat = pipe temperature – saturated evaporating temperature (normal 5 to 10 K)

Subcooling is measured here



Subcooling = saturated condensing temperature – liquid line temperature (normal 3 to 8 K)

Superheat tells you the evaporator has boiled off every drop of liquid before the compressor. Measure the suction line temperature near the evaporator outlet, read the suction pressure, convert that to a saturation temperature, and subtract. Normal is 5 to 10 K. High superheat means a starved evaporator (low charge or restriction). Low or zero superheat means liquid is heading for the compressor.

Subcooling tells you a solid column of liquid is reaching the expansion device. Measure the liquid line temperature at the condenser outlet, convert the discharge pressure to a saturation temperature, and subtract the other way. Normal is 3 to 8 K. Low subcooling with bubbles in the sight glass means low charge. High subcooling with high head pressure means overcharge.

On zeotropic blends use the **dew point** for superheat and the **bubble point** for subcooling.

The parts between the big four

- **Receiver:** stores liquid after the condenser; used for pump-down.
- **Filter drier:** desiccant and a screen in the liquid line; traps moisture, acid and debris. Change it every time the system is opened.
- **Sight glass:** a window in the liquid line. Clear is good; bubbles mean flash gas. Many have a moisture indicator that changes colour when wet.
- **Suction accumulator:** catches liquid before the compressor.
- **HP and LP cut-outs:** pressure switches that stop the compressor on excessive head pressure or on loss of charge.
- **Reversing valve:** in a heat pump, swaps which coil is the condenser and which the evaporator.
- **Schrader valves:** service ports. The core is a service aid; the **cap** is the real seal. Always refit caps.

3 Refrigerants

Families, numbers, GWP, blends, safety classes and oils. Half the exam lives in this chapter.

The families

FAMILY	CONTAINS	EXAMPLES	OZONE (ODP)	WARMING (GWP)	REGULATED BY
CFC	Cl, F, C	R11, R12	High	Very high	ODS Regulation (banned)
HCFC	H, Cl, F, C	R22	Low but real	High (1810)	ODS Regulation (banned for service since 2015)
HFC	H, F, C	R134a, R32, R410A, R404A	Zero	High	F-Gas Regulation
HFO	H, F, C (unsaturated)	R1234yf, R1234ze	Zero	Very low (about 4)	F-Gas (listed, no quota)
Natural	Hydrocarbons, CO2, ammonia	R290, R600a, R744, R717	Zero	0 to 3	Neither, but safety standards apply

Chlorine is what harms ozone, so CFCs and HCFCs are **ozone depleting substances** under the ODS Regulation. Fluorine without chlorine does not touch ozone but traps heat, so HFCs are the **F-gases**. Exam trap: R22 is *not* an F-gas.

GWP: the numbers to learn

Global Warming Potential compares 1 kg of a gas to 1 kg of CO2 over 100 years. The regulation uses the IPCC Fourth Assessment Report (AR4) values. These five come up again and again:

REFRIGERANT	TYPE	GWP	SAFETY CLASS
R32	Pure HFC	**675**	A2L
R134a	Pure HFC	**1430**	A1
R407C	Zeotropic HFC blend (R32/R125/R134a)	**1774**	A1
R22	HCFC	**1810**	A1
R410A	Near-azeotropic HFC blend (R32/R125)	**2088**	A1
R404A	HFC blend	**3922**	A1
R1234yf	HFO	**4**	A2L
R290 propane	Hydrocarbon	**3**	A3
R744 CO2	Natural	**1**	A1
R717 ammonia	Natural	**0**	B2L

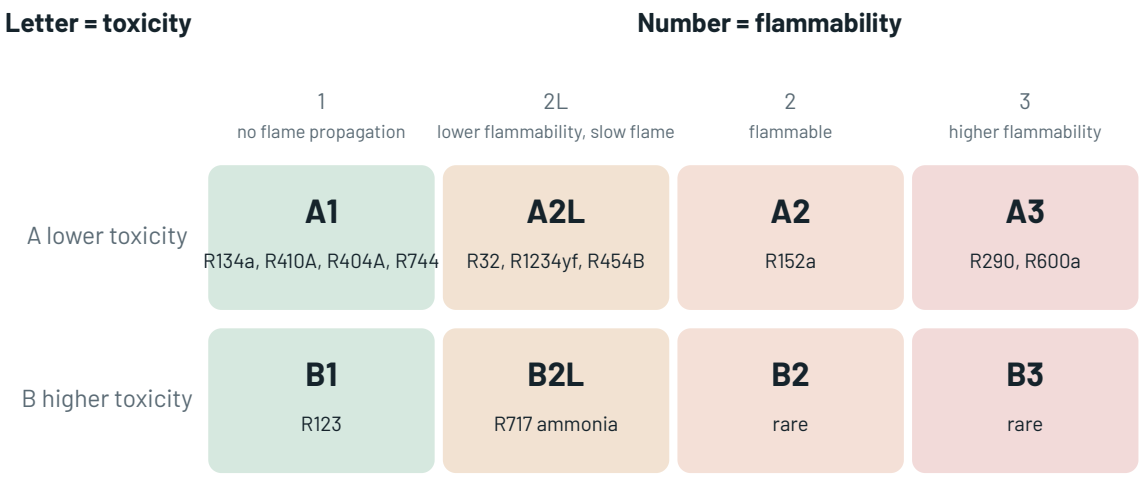
Blends and glide

Zeotropic blends (R400 series: R407C, R410A, R404A) are mixtures whose parts boil at different temperatures. At a fixed pressure they start boiling at the **bubble point** and finish at the **dew point**; the gap is

the **temperature glide**. R407C glides about 5 to 7 K; R410A so little (under 0.2 K) that it is called near-azeotropic.

Because the parts separate, zeotropic blends are always **charged as liquid** from the cylinder. Taking vapour off the top draws the lighter components first and leaves the wrong mix in both the cylinder and the system. **Azeotropic blends** (R500 series: R507A) behave as one substance and have no glide.

Safety classes



The letter is toxicity, the number is flammability. A2L refrigerants such as R32 will burn but need a concentrated leak and an ignition source, and the flame moves slowly. They still require spark-free tools, recovery equipment rated for flammables, ventilation and the charge limits in EN 378. A3 hydrocarbons are treated like LPG.

Oils

HFCs do not mix with mineral oil, so HFC systems use **polyol ester (POE)** oil. POE is strongly **hygroscopic**: it pulls moisture out of the air in minutes. Keep oil containers sealed, cap pipes, and keep systems open for the shortest time possible. Moisture plus POE plus heat makes acid, which attacks motor windings and causes **copper plating** on bearings.

Recovered, recycled, reclaimed

Recovered	Taken out of the system into a cylinder. Nothing else done.
Recycled	Basic cleaning on site or in the van: oil separation, filtering, moisture removal. May go back into equipment of the **same owner**.
Reclaimed	Reprocessed at a specialist facility to a virgin specification (AHRI 700). Can be sold and used anywhere.
Mixed	Different refrigerants in one cylinder. Cannot be reclaimed. Sent for destruction at your cost. Never mix.

4 The law: F-Gas and ODS

Who may do what, what the operator must do, and the ban dates that keep coming up.

Two regulations

The **F-Gas Regulation** (GB version of EU 517/2014) aims to cut emissions of fluorinated greenhouse gases. It sets personnel and company certification, leak checking, record keeping, labelling, the HFC phase-down and the equipment bans. The **ODS Regulation** (GB version of EC 1005/2009) controls ozone depleting substances such as R22: it banned virgin R22 for servicing from 2010 and recycled or reclaimed R22 from 1 January 2015. Systems may run on their existing R22 charge but cannot be topped up.

In Great Britain the F-Gas rules are enforced by the **Environment Agency** in England, **Natural Resources Wales** and **SEPA** in Scotland. Health and safety and pressure systems are the HSE's business.

Personnel categories

CATEGORY	LEAK CHECK (NO BREAK-IN)	RECOVERY	INSTALL, MAINTAIN, SERVICE, REPAIR, DECOMMISSION
I (2079-11)	Any size	Any size	Any size
II	Any size	Under 3 kg (6 kg hermetic)	Under 3 kg (6 kg hermetic)
III	No	Under 3 kg (6 kg hermetic)	No
IV	Any size	No	No

The 6 kg limit applies only to **hermetically sealed** systems that are labelled as such. Hermetically sealed means every refrigerant-containing part is joined by brazing, welding or a similar permanent method (capped valves and service ports allowed) and the tested leak rate is below 3 g per year.

Company certification

Certifying the engineer is not enough. Any **undertaking** (business) that installs, services, maintains, repairs or decommissions stationary refrigeration, air conditioning or heat pump equipment containing F-gas must hold a **company certificate** from an approved body such as REFCOM, Quidos or Bureau Veritas. An operator must check both certificates before letting a contractor work.

Pre-charged, non-hermetically sealed equipment (a split air conditioner, for example) may only be sold to an end user with evidence that a certified undertaking will install it.

The operator's duties

The **operator** is whoever has actual control over how the equipment works, normally the owner or the business occupying the premises. The engineer does the work but the operator carries the legal duty to:

- prevent leaks and have detected leaks repaired without undue delay, then re-checked within one month;
- have leak checks carried out at the right intervals by certified personnel;
- fit a leak detection system at 500 tonnes CO2e and above;
- keep records for five years and produce them on request;
- ensure recovery before disposal and use only certified people and companies.

Labels

Equipment containing F-gas must carry a label stating that it contains fluorinated greenhouse gases, the refrigerant designation, the charge in kg *and* in tonnes CO2e, and the GWP. If you add gas on site, update the label.

Phase-down and bans



The **HFC phase-down** works through quotas on producers and importers, measured in tonnes CO2e, stepping down year by year. It does not ban HFCs outright; it makes high-GWP gas scarce and expensive so the market moves to lower-GWP alternatives.

Alongside it sit specific bans. New stationary refrigeration with HFCs of GWP 2500 or more has been banned since 2020 (except equipment designed for below minus 50 degrees C). Since 2020 you may not service equipment with a charge of 40 tonnes CO2e or more using **virgin** HFCs of GWP 2500 or more; recycled or reclaimed gas is allowed until 2030. From 2025 single split air conditioners with under 3 kg may not use refrigerant of GWP 750 or more, which is why R32 replaced R410A.

5 CO2 equivalent and leak checking

One formula, three thresholds, and the methods the assessor wants to see.

The formula

kg of refrigerant × GWP ÷ 1000 = tonnes CO2e

Example: 12 kg of R410A → 12 × 2088 = 25,056 kg → 25.1 tonnes CO2e → leak check every 12 months

Every threshold in the regulation is in tonnes of CO2 equivalent, not kilograms, so a small charge of a high-GWP gas is caught sooner than a large charge of a low-GWP one. Work it out as kg times GWP, then divide by 1000.

SYSTEM	WORKING	TONNES CO2E	LEAK CHECK
3 kg R134a	3 × 1430 = 4290 kg	4.29	Under 5 t: no set interval
5 kg R410A	5 × 2088 = 10,440 kg	10.44	Every 12 months
1.3 kg R404A	1.3 × 3922 = 5099 kg	5.10	Every 12 months
30 kg R404A	30 × 3922 = 117,660 kg	117.7	Every 6 months
250 kg R410A	250 × 2088 = 522,000 kg	522	Every 3 months, detection system compulsory

Going the other way: the 5 tonne threshold is reached by 1.27 kg of R404A, 2.39 kg of R410A, 3.50 kg of R134a or 7.41 kg of R32.

The intervals

Charge in tonnes CO2e → how often a certified person must leak check

under 5 t	**No set interval. Still prevent and repair leaks.**
5 t or more	**Every 12 months** 24 months with a leak detection system
50 t or more	**Every 6 months** 12 months with a leak detection system
500 t or more	**Every 3 months** 6 months with detection. Detection system compulsory.

Hermetically sealed and labelled: threshold is 10 t instead of 5 t. Leak detection systems must themselves be checked every 12 mont

A fixed leak detection system doubles each interval. Labelled hermetically sealed equipment starts at 10 tonnes instead of 5. The detection system itself must be checked every 12 months. After any leak repair, re-check within one month to prove the repair worked.

Direct and indirect methods

Direct methods test the pipework itself: an electronic leak detector (sensitivity at least 5 g per year, checked every 12 months), bubble solution on joints, UV dye and lamp, or a standing pressure test with OFN. **Indirect methods** read the system: pressures, temperatures, compressor current, oil and liquid levels, the sight glass. A change in those points to a leak that you then confirm directly.

Where to look first: anywhere oil has stained. Flare nuts, brazed joints, Schrader cores, valve stems, the compressor shaft seal, capillary rubbing on a bracket. Move the detector probe slowly *under* joints, because refrigerant is heavier than air. Confirm a hit with bubbles.

On a nitrogen standing pressure test, allow for **temperature**. Pressure falls as the gas cools overnight, so compare corrected readings over a long enough period before calling it a leak.

What to do when you find one

Repair it. Topping up a known leaker instead of fixing it is an offence for the operator and the engineer. Sealants are not a repair. Record the leak, the repair, the gas added and the follow-up check in the equipment record.

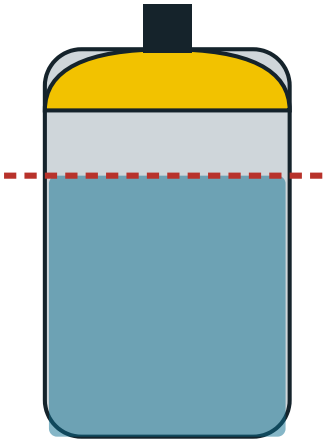
6 Recovery and disposal

Getting refrigerant out safely, keeping it clean, and moving it on legally.

When recovery is required

Before any work that opens the circuit, before decommissioning, and before equipment is scrapped. Deliberate venting is prohibited, and every technically and economically feasible step must be taken to avoid accidental release. Only a certified person may recover.

The recovery cylinder



80% of rated capacity, by WEIGHT, on scales

Above this a warm cylinder can become hydraulically full and burst.

Grey body, yellow shoulder = UK recovery cylinder

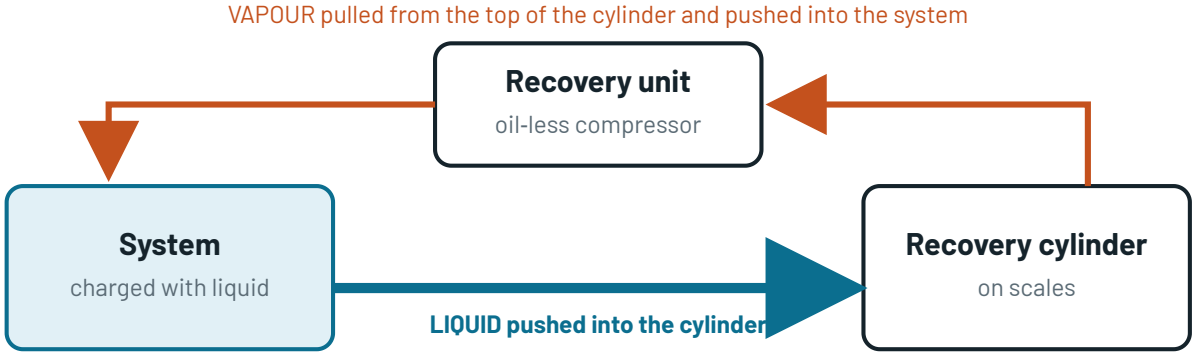
Dedicated to one refrigerant and labelled

In test date, evacuated before use, float switch as backup only

Never refill a disposable cylinder

Weigh the cylinder on scales throughout and stop at **80% of its rated capacity**. Liquid expands when warm; an overfilled cylinder can become hydraulically full and burst. The float switch on the cylinder is a backup, not the method. Check the test date, evacuate the cylinder before use so it holds no air, and dedicate it to one refrigerant.

Methods



Fast for large charges. Finish with normal vapour recovery to pull the system below atmospheric pressure.

Vapour recovery through the suction port is simple but slow. **Liquid recovery** from the liquid line or receiver moves far more refrigerant per minute. **Push-pull** uses the recovery unit to pull vapour off the cylinder and

push it into the system, driving liquid out into the cylinder; it suits large charges. Whichever you use, finish with vapour recovery until the system is below atmospheric pressure (a slight vacuum, as the recovery unit and regulation require) so almost nothing escapes when you open it. Break that vacuum with OFN before brazing.

For A2L or A3 refrigerants the recovery machine must be rated for flammables, the area ventilated and ignition sources removed.

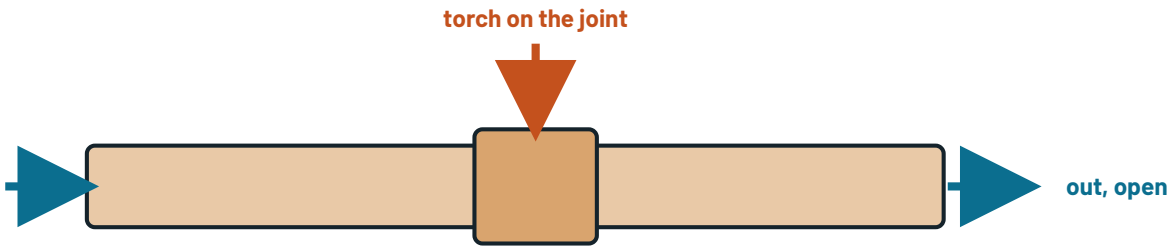
Where it goes

Recovered refrigerant and used compressor oil are **hazardous waste**. They travel with a **registered waste carrier** to a permitted facility under a **hazardous waste consignment note**, which both parties keep for three years. Recycled gas may go back into the same owner's equipment; anything else must be reclaimed by a licensed facility or destroyed. Never put recovered gas into a disposable cylinder, and never pour oil down a drain.

7 Installation and commissioning

Test, evacuate, charge, run. In that order, every time.

Brazing with a nitrogen purge



OFN in, gentle flow

through a regulator, barely felt on the back of the hand

Nitrogen displaces the air, so no black copper-oxide scale forms inside the pipe to block driers and expansion valves.

Heat copper in air and black copper-oxide scale forms inside the pipe. It breaks off and blocks driers, strainers and expansion valves. A gentle flow of **oxygen-free nitrogen** through the pipe while brazing displaces the air and keeps the bore clean. Copper-phosphorus rods are self-fluxing on copper to copper; copper to brass or steel needs a high-silver rod with flux, and the residue must be cleaned off.

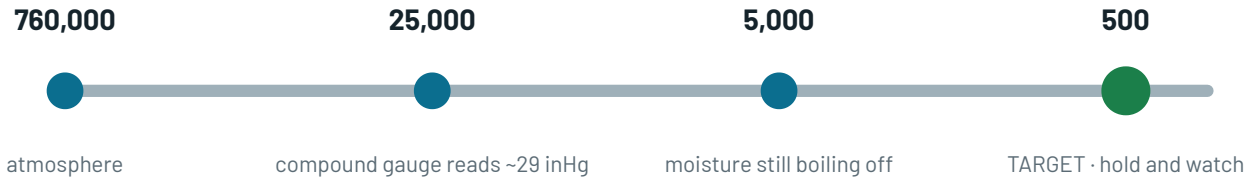
Never braze on a charged system. Recover, purge with OFN, confirm no pressure, clear flammables, have an extinguisher and a hot work permit, and keep a fire watch afterwards.

Pressure testing

Use **oxygen-free nitrogen through a regulator**. Never oxygen (explodes with oil), never compressed air (moisture, and an explosive mix with oil), never refrigerant alone. The **strength test** at 1.1 times the maximum allowable pressure (PS) proves the pipework will not burst. The **tightness test** at PS, held long enough and corrected for temperature, proves it does not leak. Both before any refrigerant goes in.

Evacuation

Vacuum in microns (absolute pressure). Lower is deeper.



Vacuum rises fast after isolating the pump = leak. Creeps up and stalls = moisture still present. Stays put = dry and tight.

Evacuating removes **air** (non-condensables that raise head pressure and waste energy) and **moisture** (which becomes acid and freezes at the expansion device). Under a deep vacuum water boils at room temperature and is pumped away. Use a two-stage pump, remove Schrader cores with a core tool to speed things up, and

measure with an **electronic micron gauge at the system**, never the compound gauge. Pull below **500 microns**, isolate the pump, and watch. A fast rise is a leak; a slow creep that stalls is moisture; steady is good.

On a wet system use **triple evacuation**: pull a vacuum, break it with OFN, pull again, break again, pull a third time. Change the vacuum pump oil often, because the moisture ends up in it and limits the vacuum the pump can reach.

Charging

Charge **by weight** on electronic scales: the nameplate charge plus the manufacturer's allowance for extra pipe length. Zeotropic blends go in as **liquid**. With the system off, liquid can go into the liquid line; with it running, meter liquid slowly into the suction so it flashes before the compressor. Never slug liquid into a running compressor. Then confirm with superheat and subcooling, not the other way round.

> **Commissioning order:** strength test, tightness test, evacuate, charge by weight, run, check superheat and subcooling, label, record.

8 Reading the symptoms

Four faults account for most call-outs. Learn their fingerprints.

Symptom table

FAULT	SUCTION PRESSURE	HEAD PRESSURE	SUPERHEAT	SUBCOOLING	SIGHT GLASS	OTHER SIGNS
Low charge	Low	Low	High	Low	Bubbles	Ice at TEV inlet, poor capacity, oil stains at the leak
Overcharge	High	High	Low	High	Clear	High compressor current, liquid floodback, warm suction line not cold
Non-condensables (air)	Normal	High, above PT chart for the liquid temperature	Normal	High-looking	Clear	Poor evacuation history, high discharge temperature
Restriction (blocked drier or TEV)	Low	Low to normal	High	Normal to high	Bubbles after the restriction	Temperature drop across the drier, frost after the blockage
Dirty condenser or fan failure	High	High	Normal	Normal to high	Clear	HP cut-out trips, condenser hot to touch

Low charge and a restriction look alike on the gauges. The difference: a low charge gives low subcooling; a restriction leaves subcooling normal and shows a temperature change across the blocked component. Find the leak before adding gas.

Moisture

A wet system shows a colour change in the moisture indicator, intermittent cooling as ice forms and clears at the expansion device, acidic oil and eventually a failed compressor. Fit a new drier, evacuate properly, and find out how the moisture got in.

9 Health and safety

Refrigerant is cold, heavy, under pressure, sometimes flammable, and poisonous in a flame.

The hazards

- **Asphyxiation.** Most refrigerants are heavier than air and odourless. A leak in a plant room, pit or cold store displaces oxygen at low level. Ventilate, use an oxygen monitor or fixed detection, never work alone in a suspected leak.
- **Frostbite.** Liquid refrigerant flashes off well below 0 degrees C and freezes skin and eyes. Gloves and eye protection whenever you connect hoses, charge or recover.
- **Toxic decomposition.** HFCs in a flame or on a hot surface break down to **hydrogen fluoride** and other corrosive gases. No brazing on charged systems, no smoking near leaks, good ventilation.
- **Pressure.** Never heat a cylinder with a flame; use a heater belt or warm water under 40 degrees C. Never use oxygen or compressed air for testing. Keep cylinders upright, secured, capped, cool and ventilated, and carry them under ADR rules.
- **Flammability.** A2L and A3 refrigerants need ignition sources removed, spark-free tools, rated recovery equipment and the charge limits of EN 378.
- **Ammonia (R717).** B2L: toxic and mildly flammable, attacks copper, but its smell gives early warning.

The paperwork that protects you

COSHH requires an assessment of every substance you use, based on its **safety data sheet**. The **Pressure Systems Safety Regulations 2000** exempt refrigeration systems with compressor motor power of 25 kW or less; above that a written scheme of examination and periodic inspection are required. Brazing on site needs a **hot work permit**, an extinguisher to hand and a fire watch after the work.

10 Records

Five years, both parties, and exactly what goes in.

What to record

For every piece of equipment subject to leak checks (5 tonnes CO2e or more, 10 for labelled hermetic), the record must show:

- the type and quantity of F-gas installed;
- quantities added during installation, maintenance or because of leaks;
- whether any gas used was recycled or reclaimed, and the name and address of the facility;
- the quantity recovered;
- the identity of the undertaking that did the work;
- the dates and results of leak checks;
- on decommissioning, the measures taken to recover and dispose of the refrigerant.

Both the **operator** and the **contractor** keep the record for at least **five years**. Hazardous waste consignment notes are kept for **three years**. Update the equipment label whenever the charge changes.

11 Practice questions

Thirty exam-style questions. Mark yourself with the key on the next pages, then read the explanations for anything you missed.

1.

REGULATIONS & CERTIFICATION

What is the main purpose of the F-Gas Regulation?

- A To reduce emissions of fluorinated greenhouse gases and limit their use
- B To protect the ozone layer from chlorine-based refrigerants
- C To set minimum energy efficiency ratings for air conditioning
- D To control the price of refrigerant cylinders

2.

REGULATIONS & CERTIFICATION

A Category IV certificate permits which activity?

- A Brazing of refrigerant pipework
- B Leak checking only, without breaking into the refrigerant circuit
- C Recovery from systems under 3 kg
- D Installation of small split systems

3.

REGULATIONS & CERTIFICATION

How does the HFC phase-down work?

- A All HFC refrigerants are banned from a single date
- B Only refrigerants with a GWP above 2500 are restricted
- C A quota system limits the amount of HFC, measured in CO2 equivalent, that producers and importers may place on the market each year
- D Each contractor is allowed a fixed number of kilograms per year

4.

REGULATIONS & CERTIFICATION

What must the label on equipment containing F-gas state?

- A Only the refrigerant name
- B The installer's phone number and the date of installation
- C The energy rating of the equipment
- D That it contains F-gas, the refrigerant designation, the charge in kg and its CO2 equivalent, and the GWP

5.

REGULATIONS & CERTIFICATION

Which of these best describes a hermetically sealed system under the F-Gas Regulation?

- A All refrigerant-containing parts are made tight by welding, brazing or a similar permanent connection, and the tested leak rate is below 3 g per year
- B Any system with a sealed compressor
- C A system that has never been opened since manufacture
- D A system with no service ports

6.

REFRIGERANTS & GWP

What is the Global Warming Potential (GWP) of carbon dioxide (R744)?

- A 10
- B 1
- C 0
- D 100

7.

REFRIGERANTS & GWP

What is the GWP of R32?

- A 2088
- B 4
- C 675
- D 1430

8.

REFRIGERANTS & GWP

Which of these is an HFO refrigerant?

- A R410A
- B R290
- C R717
- D R1234yf

9.

REFRIGERANTS & GWP

What is 'temperature glide'?

A The difference between the bubble point and dew point of a zeotropic blend at a given pressure

B The drop in temperature across the expansion valve

C The rise in temperature across the compressor

D The change in cylinder temperature while charging

10.

REFRIGERANTS & GWP

Why must different refrigerants never be mixed in a system or a recovery cylinder?

A Because it voids the cylinder warranty

B Mixed refrigerant cannot be reclaimed, must be sent for destruction, and gives unpredictable pressures and performance

C It is permitted as long as the GWPs are similar

D Because the blend will become flammable

11.

LEAK CHECKING

Equipment containing 5 tonnes CO2e or more of F-gas (but less than 50) must be leak checked at least every:

A 3 months

B 24 months in all cases

C 12 months (24 months if a fixed leak detection system is fitted)

D 6 months

12.

LEAK CHECKING

After a leak has been repaired, the equipment must be checked again to confirm the repair worked within:

A 1 week

B 6 months

C 12 months

D 1 month

13.

LEAK CHECKING

A system has 12 kg of R410A and no fixed leak detection. How often must it be leak checked?

A Every 12 months (12 x 2088 = 25 tonnes CO2e)

B Every 6 months

C Every 3 months

D No check required

14.

LEAK CHECKING

A nitrogen standing pressure test shows a small drop in pressure overnight. What must you consider before declaring a leak?

A Nothing. Any drop is a leak

B The change in ambient temperature, since pressure falls as the gas cools

C Whether the nitrogen has leaked through the pipe walls

D Whether the GWP has changed

15.

RECOVERY & DISPOSAL

A recovery cylinder must never be filled beyond what proportion of its capacity?

A 95%

B 50%

C 80% by weight, checked on scales

D 100%

16.

RECOVERY & DISPOSAL

When must refrigerant be recovered from a system?

A Only when the equipment is scrapped

B Only when the charge exceeds 3 kg

C Never, if the system is under 5 tonnes CO2e

D Before any work that opens the circuit, before decommissioning and before disposal of the equipment

17.

RECOVERY & DISPOSAL

What does the float switch on a recovery cylinder do?

A Stops the recovery machine when the cylinder reaches 80% full

B Indicates the refrigerant type

C Vents excess pressure

D Measures the vacuum level

18.

INSTALLATION & COMMISSIONING

Why is oxygen-free nitrogen purged through the pipework while brazing?

A To make the brazing rod flow more easily

B To stop copper oxide scale forming inside the pipe, which would contaminate the system and block filters and expansion valves

C To cool the joint faster

D To stop the flame blowing out

19.

INSTALLATION & COMMISSIONING

To what vacuum level should a system be evacuated before charging?

A Below 1 bar

B Below 5000 microns

C Below 500 microns (about 0.67 mbar), then held to check it does not rise

D Below 29 inches on the manifold gauge

20.

INSTALLATION & COMMISSIONING

What is the correct order of commissioning tasks on a new split system?

A Charge, evacuate, pressure test, run

B Evacuate, pressure test, charge, run

C Run, charge, pressure test, evacuate

D Strength and tightness test with OFN, evacuate, charge by weight, run and check superheat and subcooling

21.

INSTALLATION & COMMISSIONING

Which brazing rod can be used on copper-to-copper joints without flux?

A Silver-copper-phosphorus alloy (e.g. 2% to 15% silver)

B High-silver solder with flux is always needed

C Soft lead solder

D Aluminium rod

22.

SYSTEM OPERATION

In a refrigeration system, which component absorbs heat and makes the refrigerant cold?

A The receiver

B The evaporator, where low-pressure liquid boils and absorbs latent heat

C The condenser

D The compressor

23.

SYSTEM OPERATION

What is superheat and where is it measured?

A The liquid temperature below saturation at the condenser outlet

B The difference between room and evaporator temperature

C The temperature of the vapour above its saturation temperature, measured at the evaporator outlet or suction line

D The temperature of the compressor discharge

24.

SYSTEM OPERATION

What is latent heat?

A Heat that changes the temperature of a substance

B Heat produced by the compressor motor

C Heat lost through pipe insulation

D Heat added or removed during a change of state at constant temperature

25.

SYSTEM OPERATION

Which of these components are on the high side of the system?

A Condenser, receiver, liquid line, filter drier and sight glass

B Evaporator and suction line

C Suction accumulator

D Compressor suction valve

26.

HEALTH & SAFETY

What is the main hazard of a large refrigerant leak in a plant room or cold store?

A Electric shock

B Asphyxiation, because the vapour is heavier than air and displaces oxygen at low level

C Fire from all refrigerants

D Skin irritation

27.

HEALTH & SAFETY

Before brazing on a system that has contained refrigerant you should:

A Leave a small charge in to show up leaks

B Run the compressor to clear the pipe

C Recover the refrigerant, purge with OFN, confirm no residual pressure, remove flammables and have an extinguisher and hot work permit in place

D Open a window and start brazing

28.

HEALTH & SAFETY

Which properties make ammonia (R717) hazardous?

A It is odourless and heavier than air

B It has a very high GWP

C It is harmless but expensive

D It is toxic, mildly flammable (B2L) and attacks copper, but its strong smell gives early warning

29.

RECORDS & OPERATOR DUTIES

How long must F-gas records for equipment be kept?

A At least 5 years, by both the operator and the contractor who did the work

B 1 year

C 3 months

D Until the next leak check

30.

RECORDS & OPERATOR DUTIES

When refrigerant is added to a system on site, what must the engineer do to the equipment label?

A Write the date only

B Update it with the new total charge in kg and CO2e

C Nothing, the original label stays

D Remove it

11 Answers and explanations

1 **A.** To reduce emissions of fluorinated greenhouse gases and limit their use

The F-Gas Regulation (EU 517/2014, kept in GB law after Brexit) exists to cut emissions of fluorinated greenhouse gases such as HFCs. Ozone protection is the job of the separate ODS Regulation.

2 **B.** Leak checking only, without breaking into the refrigerant circuit

Category IV is leak checking only, and only by methods that do not break into the circuit. Category III is recovery from small systems.

3 **C.** A quota system limits the amount of HFC, measured in CO2 equivalent, that producers and importers may place on the market each year

The phase-down works through quotas on producers and importers, measured in tonnes CO2e. The quota steps down over time, which pushes the market towards lower-GWP refrigerants and makes high-GWP gas scarce and expensive.

4 **D.** That it contains F-gas, the refrigerant designation, the charge in kg and its CO2 equivalent, and the GWP

Labels must say the equipment contains fluorinated greenhouse gases, give the refrigerant designation (e.g. R410A), the quantity in kg and tonnes CO2e, and the GWP. If gas is added on site the label must be updated.

5 **A.** All refrigerant-containing parts are made tight by welding, brazing or a similar permanent connection, and the tested leak rate is below 3 g per year

The definition requires permanent joints (brazed or welded, though capped valves and service ports are allowed) and a leak rate under 3 g per year at a quarter of the maximum allowable pressure. Such systems are given relaxed leak check thresholds when labelled as hermetically sealed.

6 **B.** 1

GWP is measured relative to CO2 over 100 years, so CO2 has a GWP of 1 by definition. Everything else is compared against it.

7 **C.** 675

R32 has a GWP of 675, roughly a third of R410A, which is why it replaced R410A in small splits. It is A2L (mildly flammable).

8 **D.** R1234yf

R1234yf and R1234ze are hydrofluoro-olefins (HFOs). They have very low GWP (around 4 or less) but are A2L mildly flammable. R410A is an HFC blend, R290 is propane and R717 is ammonia.

9 **A.** The difference between the bubble point and dew point of a zeotropic blend at a given pressure

Zeotropic blends do not boil or condense at one fixed temperature. At a given pressure they start to boil at the bubble point and finish at the dew point. That spread is the glide. Use the dew point for superheat and the bubble point for subcooling.

10 **B.** Mixed refrigerant cannot be reclaimed, must be sent for destruction, and gives unpredictable pressures and performance

Mixed gas has no pressure-temperature data, can damage the system, and cannot be separated economically. It has to be destroyed at cost. Always identify the refrigerant before recovery and use a dedicated cylinder.

11 **C.** 12 months (24 months if a fixed leak detection system is fitted)

The intervals are: 5 tonnes or more every 12 months, 50 tonnes or more every 6 months, 500 tonnes or more every 3 months. A fixed leak detection system doubles the allowed interval in each band.

12 **D.** 1 month

The regulation requires a follow-up leak check within one month of a repair to verify it was effective. Ideally the repaired joint is also tested before recharging.

13 **A.** Every 12 months (12 x 2088 = 25 tonnes CO2e)

12 kg x 2088 = 25,056 kg = 25.06 tonnes CO2e. That is in the 5 to 50 tonne band, so every 12 months.

14 **B.** The change in ambient temperature, since pressure falls as the gas cools

Gas pressure changes with temperature, so a reading taken cold in the morning will be lower than one taken warm. Compare temperature-corrected readings over a long enough period before concluding.

15 **C.** 80% by weight, checked on scales

Liquid expands when warmed. A cylinder filled above 80% can become hydraulically full and rupture. Weigh the cylinder on scales during recovery and stop at 80% of its rated capacity (or rely on the float switch as a backup only).

16 **D.** Before any work that opens the circuit, before decommissioning and before disposal of the equipment

Recovery is required whenever the refrigerant would otherwise be released: opening the circuit for repair, decommissioning, and before equipment is scrapped. A certified person must do it.

17 **A.** Stops the recovery machine when the cylinder reaches 80% full

The float switch is a backup to prevent overfilling. It should not replace weighing the cylinder on scales.

18 **B.** To stop copper oxide scale forming inside the pipe, which would contaminate the system and block filters and expansion valves

Heating copper in air forms black oxide scale on the inside of the pipe. Scale breaks off and blocks driers, strainers and expansion valves. A gentle OFN purge displaces the air and keeps the pipe clean.

19 **C.** Below 500 microns (about 0.67 mbar), then held to check it does not rise

Industry practice is to pull below 500 microns and hold. If the vacuum rises quickly there is a leak; if it creeps up slowly and stalls there is still moisture boiling off. A manifold compound gauge is not accurate enough.

20 **D.** Strength and tightness test with OFN, evacuate, charge by weight, run and check superheat and subcooling

Pressure test first (leaks are found before the gas goes in), then evacuate to remove air and moisture, then weigh in the charge, then run the system and verify performance.

21 **A.** Silver-copper-phosphorus alloy (e.g. 2% to 15% silver)

Copper-phosphorus rods are self-fluxing on copper. Joining copper to brass or steel needs a high-silver alloy with a suitable flux, and the flux residue must be cleaned off afterwards.

22 **B.** The evaporator, where low-pressure liquid boils and absorbs latent heat

After the expansion device the refrigerant is a cold, low-pressure liquid. In the evaporator it boils, taking latent heat from the air or product. The condenser does the opposite, rejecting heat as the vapour turns back to liquid.

23 **C.** The temperature of the vapour above its saturation temperature, measured at the evaporator outlet or suction line

Superheat = actual suction line temperature minus the saturated evaporating temperature (found from the suction pressure on a PT chart). It proves all the liquid has boiled before the compressor. Typically 5 to 10 K.

24 **D.** Heat added or removed during a change of state at constant temperature

Latent heat is absorbed when a liquid boils and released when a vapour condenses, with no change in temperature. Sensible heat changes temperature. Refrigeration relies on latent heat because it moves far more energy per kilogram.

25 **A.** Condenser, receiver, liquid line, filter drier and sight glass

The high side runs from the compressor discharge through the condenser, receiver, liquid line, drier and sight glass to the expansion device. After the expansion device everything is low side until the compressor suction.

26 **B.** Asphyxiation, because the vapour is heavier than air and displaces oxygen at low level

Most refrigerants are heavier than air and odourless. They collect in low areas and displace oxygen. Ventilate, use an oxygen monitor or fixed detection, and never enter a suspected leak area alone.

27 **C.** Recover the refrigerant, purge with OFN, confirm no residual pressure, remove flammables and have an extinguisher and hot work permit in place

Brazing a pressurised line can blow molten metal out of the joint and decompose the refrigerant into toxic gas. Recover, make safe, purge with OFN and follow hot work procedures.

28 **D.** It is toxic, mildly flammable (B2L) and attacks copper, but its strong smell gives early warning

Ammonia is classified B2L. It is toxic at low concentrations but its pungent smell means leaks are noticed quickly. It cannot be used with copper pipe or windings, so ammonia systems use steel.

29 **A.** At least 5 years, by both the operator and the contractor who did the work

Both the operator and the undertaking (contractor) must keep records for at least five years and make them available to the regulator on request.

30 **B.** Update it with the new total charge in kg and CO2e

The label must reflect the actual quantity of F-gas in the equipment, so after adding gas the label (or an additional label) must be updated with the new charge.

12 Numbers to memorise

If you learn one page, learn this one.

3 kg / 6 kg hermetic	Charge limits for Category II and III work
3 g per year	Max leak rate to count as hermetically sealed
5 / 50 / 500 t CO2e	Leak check every 12 / 6 / 3 months (24 / 12 / 6 with detection)
10 t CO2e	Leak check threshold for labelled hermetic equipment
500 t CO2e	Fixed leak detection system compulsory
1 month	Re-check after a leak repair
12 months	Check interval for leak detection systems and electronic detectors
5 g per year	Minimum sensitivity of an electronic leak detector
5 years	F-gas records kept by operator and contractor
3 years	Hazardous waste consignment notes kept
80%	Maximum recovery cylinder fill, by weight
500 microns	Target vacuum before charging
1.1 × PS	Strength test pressure with OFN
5 to 10 K / 3 to 8 K	Typical superheat / subcooling
40 °C	Maximum for warming a cylinder (no flames)
25 kW	Compressor motor power above which PSSR written scheme applies
2010 / 2015	Virgin R22 banned / recycled R22 banned
2020	No new stationary refrigeration ≥2500 GWP; service ban on virgin ≥2500 GWP in ≥40 t CO2e systems
2025	Single splits under 3 kg: no refrigerant ≥750 GWP
2030	Recycled/reclaimed ≥2500 GWP also banned for service
675 / 1430 / 1774 / 1810 / 2088 / 3922	GWP of R32 / R134a / R407C / R22 / R410A / R404A
1 / 0	GWP of CO2 / ODP of all HFCs and HFOs

13 Glossary

The trade's words, in plain English. The same 94 terms are tappable in the practice app.

A1

Safety class: lower toxicity and no flame propagation. Examples R134a, R410A, R404A. Still an asphyxiation and frostbite hazard.

A2L

Safety class: A = lower toxicity, 2L = lower flammability with a slow burning velocity. Examples R32, R1234yf, R454B. Needs ignition sources removed and equipment rated for A2L.

A3

Safety class: lower toxicity but higher flammability. Hydrocarbons such as R290 (propane) and R600a (isobutane). Strict charge limits and spark-free equipment.

Absolute pressure (bar absolute, bar a)

Pressure measured from a perfect vacuum (zero). Absolute pressure = gauge pressure + atmospheric pressure (about 1.013 bar). Pressure-temperature charts and vacuum levels use absolute pressure, so a gauge reading of 0 bar is about 1 bar absolute.

ADR

The European agreement on carrying dangerous goods by road. Refrigerant cylinders in a vehicle need correct labelling, securing, documents and in some cases driver training.

Ambient (ambient temperature)

The temperature of the surrounding air. Head pressure rises with ambient on an air-cooled condenser, and nitrogen test pressures change with it.

AR4 (IPCC AR4)

The IPCC Fourth Assessment Report (2007). Its 100-year GWP values are the ones written into the F-Gas Regulation and used for CO2e calculations.

Asphyxiation

Suffocation from lack of oxygen. Refrigerant vapour is heavier than air and odourless, so a leak in a plant room or pit displaces the oxygen at low level without warning.

Azeotropic (azeotrope)

A blend that behaves like a single refrigerant, boiling and condensing at one temperature with no glide. Numbered in the R500 series (R507A, R502).

B2L

Safety class: higher toxicity and lower flammability. Ammonia (R717).

Brazing (braze, brazed)

Joining copper pipe with a filler rod that melts above 450 degrees C (silver or copper-phosphorus alloy), using an oxy-acetylene or similar torch. Not soft soldering.

Bubble point

For a blend at a given pressure, the temperature at which the first bubble of vapour forms when heating liquid. Use it when calculating subcooling.

CFC (CFCs, chlorofluorocarbon)

Chlorofluorocarbon. The original refrigerants such as R12 and R11, with very high ODP and GWP. Long banned.

CO2 equivalent (CO2e, tonnes CO2e)

The charge of refrigerant expressed as the amount of CO2 with the same warming effect. kg of refrigerant x GWP = kg CO2e; divide by 1000 for tonnes. Leak check and record thresholds (5, 50 and 500 tonnes) use this figure.

Company certificate (company certification, REFCOM)

A certificate held by a business (not an individual) from an approved body such as REFCOM, Quidos or Bureau Veritas, confirming it has certified staff, the right tools and procedures. Required to carry out F-gas work.

Compound gauge

The blue low-side gauge on a manifold. It reads both vacuum (below atmospheric) and pressure, hence compound. Not accurate enough to measure a deep vacuum.

Compressor

The pump of the system. Draws in low-pressure vapour from the evaporator and discharges it at high pressure and temperature so it can condense.

Condenser

The heat exchanger where high-pressure vapour from the compressor gives up its heat and condenses to liquid. This is the hot coil, outdoors on an air conditioner.

Consignment note

The document that travels with hazardous waste from producer to carrier to disposal site. Copies are kept for three years.

Copper plating

Copper dissolved by acid in a wet system deposits itself on hot steel bearing surfaces inside the compressor, causing wear and seizure. A sign of moisture contamination.

COSHH

Control of Substances Hazardous to Health Regulations. Requires employers to assess and control the risks from chemicals, including refrigerants, using information from the safety data sheet.

Decommissioning (decommission, decommissioned)

Taking equipment permanently out of service. The refrigerant must be recovered by a certified person and the recovery and disposal recorded.

Dew point

For a blend at a given pressure, the temperature at which the last drop of liquid evaporates. Use it when calculating superheat.

Direct method (direct methods, direct leak)

Leak checking the pipework itself: electronic detector, bubble solution, UV dye, or a nitrogen standing pressure test.

EN 378

The European standard for the safety and environmental requirements of refrigerating systems and heat pumps. Covers pressure testing, charge limits for flammable refrigerants and plant room requirements.

Evacuation (evacuate, evacuating)

Pulling a vacuum on the system with a vacuum pump to remove air (non-condensables) and boil off moisture before charging. Target below 500 microns, then hold to check it does not rise.

Evaporator

The heat exchanger where low-pressure liquid refrigerant boils and absorbs heat from the air, water or product. This is the cold coil, indoors on an air conditioner.

Expansion device (expansion valve, capillary)

The component (TEV, electronic expansion valve or capillary tube) that drops the refrigerant pressure from the high side to the low side, so the liquid can boil at a low temperature in the evaporator.

F-gas (F-gases, fluorinated greenhouse gas)

Fluorinated greenhouse gas: HFCs, PFCs and SF6. They do not harm the ozone layer but have high GWPs. The F-Gas Regulation controls their use, leak checking, recovery and records.

Filter drier (drier, filter-drier)

A component in the liquid line containing desiccant to absorb moisture and acid, with a screen to trap particles. Replace it whenever the system is opened.

Flash gas

Vapour formed when liquid refrigerant suddenly drops in pressure, for example in the expansion device or in a liquid line with too little subcooling. Bubbles in the sight glass are flash gas.

Floodback (liquid floodback, slugging)

Liquid refrigerant reaching the compressor. Liquid cannot be compressed, so it damages valves and pistons and washes oil off the bearings.

Flux

A paste that cleans the joint and stops oxides forming so the filler alloy flows. Needed for copper to brass or steel; not needed with copper-phosphorus rods on copper to copper. Residue must be cleaned off.

Frostbite (cold burns)

Freezing injury to skin or eyes caused by liquid refrigerant flashing off at well below 0 degrees C. Prevented by gloves and eye protection.

Gauge pressure (bar gauge, bar g)

Pressure measured relative to the atmosphere around you, which is what a normal manifold gauge shows. 0 bar gauge means the same as atmospheric pressure. Gauge pressure = absolute pressure minus about 1.013 bar.

GWP (global warming potential)

Global Warming Potential. How much heat 1 kg of a gas traps over 100 years compared with 1 kg of CO2, which is 1. The F-Gas Regulation uses the IPCC AR4 values: R32 = 675, R134a = 1430, R407C = 1774, R410A = 2088, R404A = 3922.

Hazardous waste

Waste that is harmful to people or the environment, which includes recovered refrigerant and used refrigeration oil. Must be moved by a registered carrier to a permitted site with a consignment note.

HCFC (HCFCs, hydrochlorofluorocarbon)

Hydrochlorofluorocarbon. Contains chlorine so it depletes ozone, e.g. R22. Banned for servicing since 2015 under the ODS Regulation.

Head pressure (discharge pressure, condensing pressure)

The pressure on the high side, between the compressor discharge and the expansion device. High head pressure points to a dirty condenser, fan failure, overcharge or non-condensables.

Hermetically sealed (hermetic)

A system where every refrigerant-containing part is joined by brazing, welding or a similar permanent joint, and the tested leak rate is below 3 g per year. Service ports and capped valves are allowed. When labelled as such, the leak check threshold rises from 5 to 10 tonnes CO2e.

HFC (HFCs, hydrofluorocarbon)

Hydrofluorocarbon. Synthetic refrigerants containing hydrogen, fluorine and carbon but no chlorine, e.g. R134a, R32, R410A, R404A. Zero ODP but high GWP. These are the main F-gases.

HFO (HFOs, hydrofluoro-olefin)

Hydrofluoro-olefin. Newer synthetic refrigerants such as R1234yf and R1234ze with very low GWP (about 4 or less). Usually A2L, mildly flammable.

High side

The high-pressure part of the circuit: compressor discharge, condenser, receiver, liquid line, drier and sight glass up to the expansion device. The red gauge reads it.

Hot work permit (hot work)

A site authorisation for brazing, welding or grinding. Sets out precautions: removing flammables, extinguisher on hand, fire watch during and after the work.

HP cut-out (high pressure cut-out, high pressure (HP) cut-out)

A pressure switch that stops the compressor when discharge pressure rises too high.

Hydrogen fluoride

A highly toxic and corrosive gas produced when HFC refrigerants break down in a flame or on a very hot surface. The reason you never braze on a charged system.

Hygroscopic

Absorbs moisture from the air. POE oil is strongly hygroscopic, which is why moisture control matters so much on HFC systems.

Indirect method (indirect methods, indirect leak)

Leak checking by looking at system behaviour: pressures, temperatures, compressor current, oil and liquid levels, sight glass. A change suggests a leak which is then confirmed by a direct method.

Latent heat

Heat absorbed or released while a substance changes state (liquid to vapour or back) with no change in temperature. The evaporator absorbs latent heat as refrigerant boils; the condenser rejects it as vapour condenses.

Leak detection system (fixed leak detection, leak detection systems)

A fixed, calibrated device that monitors for refrigerant in the air or system and alerts the operator. Mandatory at 500 tonnes CO2e and above; where fitted it doubles the allowed leak check interval. Must be checked every 12 months.

Low side

The low-pressure part of the circuit: from the expansion device through the evaporator and suction line to the compressor inlet. The blue compound gauge reads it.

LP cut-out (low pressure cut-out, low pressure (LP) cut-out)

A pressure switch that stops the compressor when suction pressure falls too low, protecting it on loss of charge or a blockage.

Maximum allowable pressure (PS)

PS, the maximum pressure the system or component is designed and rated for, stated by the manufacturer. Strength tests are done at 1.1 x PS and tightness tests at PS.

Micron (microns)

A unit of very low absolute pressure used for vacuum work. 1000 microns = 1 torr = 1.33 mbar. 500 microns is about 0.67 mbar absolute. Atmospheric pressure is about 760,000 microns.

Mineral oil

Traditional compressor oil used with CFC and HCFC refrigerants such as R22. Does not mix with HFCs, so HFC systems use POE oil.

Natural refrigerant (natural refrigerants)

Refrigerants that occur in nature: hydrocarbons like R290 (propane) and R600a (isobutane), R744 (CO2) and R717 (ammonia). Very low GWP but each has its own hazard: flammability, high pressure or toxicity.

Non-condensables (non-condensable)

Gases such as air or nitrogen that will not condense in the condenser. They collect there, raise head pressure above the saturation pressure and reduce efficiency. Removed by proper evacuation.

ODP (ozone depletion potential, ozone depleting)

Ozone Depletion Potential. How much a gas damages the ozone layer compared with R11 (CFC-11) at 1.0. Refrigerants containing chlorine or bromine (CFCs, HCFCs) have an ODP; HFCs and HFOs have an ODP of zero.

ODS (ozone depleting substance, ozone depleting substances)

Ozone Depleting Substance. Refrigerants such as R22 (HCFC) and R12 (CFC) that contain chlorine. Controlled by the ODS Regulation, separate from the F-Gas Regulation.

OFN (oxygen-free nitrogen, oxygen free nitrogen)

Oxygen-Free Nitrogen. Dry, inert gas used for pressure testing, purging while brazing and breaking a vacuum. Always used through a regulator because the cylinder is at around 200 bar.

Operator

Under the F-Gas Regulation, the person or company with actual control over the technical functioning of the equipment, normally the owner. Responsible for leak checks, records and using certified people.

Oxide scale (copper oxide scale, copper oxide)

Black flaky copper oxide that forms inside pipe when it is brazed with air in it. It breaks off and blocks driers, strainers and expansion valves. Prevented by purging with OFN.

Personnel certificate (personnel certificates)

An individual's F-gas qualification, such as City and Guilds 2079-11 Category I. Shows the person is competent to do the activities covered by that category.

Phase-down (quota)

The stepped reduction in the amount of HFC (in tonnes CO2e) that producers and importers may place on the market each year, enforced through quotas. It makes high-GWP gas scarce and expensive.

POE oil (POE, polyol ester)

Polyol ester oil, used with HFC refrigerants. It is hygroscopic (absorbs moisture from the air), so keep containers sealed and systems open for as short a time as possible.

PSSR (Pressure Systems Safety Regulations)

Pressure Systems Safety Regulations 2000. Refrigeration systems with compressor motor power above 25 kW need a written scheme of examination and periodic inspection by a competent person.

PT chart (pressure-temperature chart, pressure temperature)

A pressure-temperature chart or slide rule showing the saturation temperature of each refrigerant at each pressure. Used to work out superheat and subcooling from gauge readings.

Pump-down (pump down)

Running the compressor with the liquid line valve closed to collect the charge in the receiver or condenser, so the low side can be opened for service without recovering the whole charge.

Push-pull (push-pull method)

A fast recovery method for large liquid charges. The recovery machine pulls vapour from the top of the cylinder and pushes it into the system, forcing liquid out of the system into the cylinder.

Receiver (liquid receiver)

A vessel after the condenser that stores liquid refrigerant and allows for changes in charge demand. Also used to hold the charge during pump-down.

Reclamation (reclaimed, reclaim)

Reprocessing recovered refrigerant at a specialist facility to meet a virgin product standard (e.g. AHRI 700) so it can be sold and used anywhere.

Recovery (recover, recovered)

Removing refrigerant from a system into a recovery cylinder using a recovery machine, so it is not released to atmosphere. Required before opening the circuit, decommissioning or scrapping.

Recovery cylinder (recovery cylinders)

A refillable pressure cylinder, grey with a yellow shoulder in the UK, dedicated to one refrigerant and in test date. Never filled beyond 80% by weight.

Recycling (recycled, recycle)

Basic cleaning of recovered refrigerant (oil separation, filtering, moisture removal) so it can be reused, normally in equipment belonging to the same owner.

Reversing valve (four-way valve)

The four-way valve in a heat pump that redirects the compressor discharge so the indoor coil becomes the condenser (heating) or the evaporator (cooling).

Safety data sheet (SDS)

The supplier's document for a substance giving its hazards, handling, storage, PPE, first aid and disposal information.

Saturation temperature (saturated evaporating temperature, saturated condensing temperature)

The temperature at which a refrigerant boils or condenses at a given pressure. Fix the pressure and you fix the temperature. A PT chart lists this relationship for each refrigerant.

Schrader valve (Schrader, Schrader core)

A spring-loaded service valve like a tyre valve, used at gauge ports. The core is for service access only; the screw-on cap with its seal is the primary leak seal.

Sensible heat

Heat that changes the temperature of something without changing its state, such as warming air or cooling liquid refrigerant below saturation.

Service ban (2020 service ban)

Since 1 January 2020, virgin HFCs with a GWP of 2500 or more may not be used to service equipment with a charge of 40 tonnes CO2e or more. Recycled or reclaimed gas is allowed until 2030.

Sight glass

A small window in the liquid line. Clear glass means a full column of liquid; bubbles mean flash gas, usually from low charge or a restriction. Many include a moisture indicator that changes colour when wet.

Strength test (strength (pressure) test, pressure test)

A test with OFN at 1.1 times the maximum allowable pressure to prove the pipework and joints will not burst. Done before the tightness test and before any refrigerant is added.

Subcooling (subcool, subcooled)

How many degrees a liquid is below its saturation temperature. Measured at the condenser outlet: saturated condensing temperature from the discharge pressure minus the actual liquid line temperature. Proves solid liquid is reaching the expansion device. Typically 3 to 8 K.

Suction accumulator (accumulator)

A vessel in the suction line that catches any liquid refrigerant before it reaches the compressor, letting it boil off safely.

Suction pressure (evaporating pressure)

The pressure on the low side, in the evaporator and suction line. Low suction pressure points to low charge, a restriction or a blocked evaporator.

Superheat (superheated)

How many degrees a vapour is above its saturation temperature. Measured at the evaporator outlet: actual suction line temperature minus the saturated evaporating temperature from the suction pressure. Proves all liquid has boiled off before the compressor. Typically 5 to 10 K.

Temperature glide (glide)

The temperature range over which a zeotropic blend boils or condenses at a fixed pressure: the difference between its bubble point and dew point. R407C has a glide of about 5 to 7 K; R410A is under 0.2 K.

TEV (thermostatic expansion valve, TXV)

Thermostatic Expansion Valve. Meters liquid into the evaporator and controls superheat using a sensing bulb clamped to the evaporator outlet.

Tightness test (standing pressure test, standing pressure)

A leak test with OFN, usually at the maximum allowable pressure, held long enough (often 24 hours) to show no pressure drop once temperature changes are allowed for.

Triple evacuation

Pull a vacuum, break it with OFN, pull again, break again, pull a third time. The dry nitrogen absorbs residual moisture, which is removed on the next pull. Used on wet or contaminated systems.

Undertaking

The regulation's word for a business or company. An undertaking carrying out F-gas work needs a company certificate, as well as certified engineers.

Virgin refrigerant (virgin)

New, unused refrigerant straight from the manufacturer, as opposed to recovered, recycled or reclaimed gas.

Written scheme of examination

A document drawn up by a competent person listing which parts of a pressure system must be inspected, how and how often, as required by PSSR.

Zeotropic (zeotrope)

A blend whose components boil at different temperatures, so it has a temperature glide. Numbered in the R400 series (R407C, R410A, R404A). Must be charged as liquid so the mix stays correct.

F-Gas Category I Study Guide. Independent revision material for the City & Guilds 2079-11 syllabus. Not affiliated with City & Guilds, HSS Training or any awarding body. Regulations change: check the current GB F-Gas and ODS Regulations before relying on a date or threshold at work.

Practice app: fgas-trainer.vercel.app